Uday kiran Reddy Buchagari | Network Engineer

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PROFESSIONAL SUMMARY

CCNA certified **Network Engineer** with 4+ years of experience supporting and optimizing enterprise and healthcare network environments, with strong expertise in **Layer 2/Layer 3 networking**, **firewall security**, and **hybrid cloud connectivity**. Hands-on experience designing and supporting **AWS networking** (VPC, VPN, subnetting) and improving operational efficiency through **Python-based automation**. Proven track record of maintaining **high availability**, reducing incident resolution time, and delivering stable, well-documented network infrastructures in production environments.

TECHNICAL SKILLS

- **Networking:** Routing & Switching (BGP, OSPF, EIGRP), VLANs, Inter-VLAN Routing, STP, HSRP, TCP/IP, LAN/WAN, MPLS, SD-WAN, QoS
- **Firewall and Security:** Cisco ASA, Palo Alto, Fortinet, Cisco ISE, VPN (Site-to-Site, Remote Access), NAT, ACLs, IPS/IDS, NAC
- **Cloud & Virtualization:** AWS VPC, Subnets, Route Tables, NAT Gateway, Internet Gateway, VPN Connectivity, Azure Networking, VMware ESXi
- **Monitoring & Analysis:** SolarWinds, Wireshark, Splunk, SNMP, NetFlow
- **Automation & Scripting:** Python (network automation, scripting), Ansible (basic automation), Bash
- **Tools:** ServiceNow, Jira, Git, Putty, SecureCRT
- **Protocols & Standards:** DHCP, DNS, NAT, IPv4/IPv6, HIPAA, PCI-DSS
- **Project & Documentation:** Visio, Confluence, Change Management, Runbooks, Network Diagrams

PROJECTS

- AWS networking (**VPC, VPN, BGP**-based routing and failover validation)
- Network monitoring (**SNMP, NetFlow, SolarWinds**) with MTTR reduction through proactive alerting
- Firewall optimization (**Cisco ASA, Palo Alto, Fortinet, VPN, NAT**)
- Routing & HA (**OSPF, EIGRP, BGP, HSRP**, failover)

PROFESSIONAL EXPERIENCE

Freyr Solutions | Princeton, New Jersey

Network Engineer | Feb 2025 – Present

- Supported enterprise network environments across multi-site infrastructure, maintaining high availability (**99.9% uptime**) through proactive **monitoring** and **incident response**
- Designed and implemented Layer 2/Layer 3 configurations, including **VLAN segmentation**, **inter-VLAN routing**, and routing protocol optimization (**OSPF, EIGRP, BGP**), improving traffic flow and convergence performance
- Deployed and optimized switching infrastructure using **Cisco Catalyst** and **Nexus** platforms, improving throughput and reducing network latency
- Supported hybrid connectivity between on-prem infrastructure and **AWS VPC** environments, configuring subnets, route tables, and secure site-to-site **VPN tunnels**
- Configured high availability solutions (**HSRP**, redundant paths, failover mechanisms), minimizing downtime during network incidents
- Managed firewall environments across **Cisco ASA, Palo Alto**, and **Fortinet**, implementing access policies, **NAT**, and **VPN** configurations to enhance security
- Developed **Python** scripts for configuration validation and operational data collection, reducing manual effort and improving accuracy
- Monitored network performance using **SNMP**, **SolarWinds**, **NetFlow**, and **Wireshark**, reducing **MTTR by 30%** approximately through proactive issue detection
- Performed regular network health checks, including interface utilization, packet drops, and routing stability to prevent potential outages
- Troubleshot complex network issues using packet analysis and routing diagnostics, restoring services within SLA timelines
- Maintained IP addressing and services (**DHCP, DNS, NAT, IPv4/IPv6**) across hybrid environments

- Created detailed network documentation including diagrams, runbooks, and configuration standards, improving operational efficiency and onboarding
- Participated in change management processes, validating configurations and ensuring controlled deployment of network changes

Citibank | New York, NY

Network Engineer | Feb 2024 – Jan 2025

- Managed network operations supporting **150+ branch environments**, maintaining **99.9% uptime** for critical banking systems
- Configured and supported Layer 2 and Layer 3 networks, including VLANs, inter-VLAN routing, STP, and routing protocols **(BGP, OSPF, EIGRP, HSRP)**
- Administered **Cisco ASA, Palo Alto, and Fortinet firewalls**, strengthening perimeter security and supporting PCI-DSS compliance
- Built and maintained secure site-to-site **VPN tunnels**, improving secure communication and reducing inter-branch latency
- Supported integration of on-prem infrastructure with **AWS VPC** environments, enabling hybrid connectivity for banking applications
- Monitored network performance using **SNMP, SolarWinds, NetFlow, Splunk**, and **Wireshark**, improving incident response time by **35%**
- Worked within **MPLS**-based WAN environments and gained basic exposure to **SD-WAN** concepts, including traffic path selection and centralized network control, improving application performance
- Applied **QoS** policies to prioritize critical financial transactions and reduce congestion
- Performed **device upgrades** and lifecycle management across network hardware, improving system stability
- Conducted network health checks and performance tuning to identify and resolve bottlenecks
- Collaborated with security and compliance teams to meet regulatory requirements
- Maintained accurate **documentation** including network diagrams, configurations, and SOPs
- Supported application migrations, ensuring stable routing and zero service disruption

Vivagene | Hyderabad, India

Network Support Engineer | Aug 2020 – July 2022

- Supported internal hospital network infrastructure, ensuring reliable connectivity across clinical and administrative environments serving **10,000+ users**
- Performed hands-on network technician tasks including **rack and stack**, device installation, patch panel management, and structured **cabling** in MDF/IDF environments
- Installed, labeled, and tested Ethernet and fiber cabling, reducing physical layer issues and improving connectivity reliability
- Troubleshot **Layer 1** and **Layer 2** issues including cable faults, port errors, and VLAN mismatches, minimizing downtime
- Configured and supported Layer 2 switching, including **VLANs**, trunking, and port configurations
- Assisted with Layer 3 routing, including **inter-VLAN routing** and basic troubleshooting of **OSPF** and **EIGRP**
- Supported configuration and maintenance of Cisco routers, switches, and **ASA** firewalls under senior engineer guidance
- Monitored network performance using **SNMP**, **SolarWinds**, and **NetFlow**, identifying issues and escalating critical incidents
- Performed routine network health checks including device status and connectivity validation
- Assisted in maintaining **VPN** connectivity for secure access to internal systems
- Participated in network upgrades including switch replacements and cabling deployments and device installations
- Maintained network documentation including diagrams and cabling records
- Participated in **on-call rotations**, assisting in incident resolution and maintaining uptime

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CERTIFICATION

- Cisco Certified Network Associate **(CCNA)**

EDUCATION

- Master of Science in Computer Information Systems - **Rivier University, Nashua, NH, USA**
- Bachelor of Science in computer science – **MLRITM, Hyderabad, India**